

Sustainable poultry farming: A concept of IoT-based poultry management system for small-scale farmers

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ORIGINAL STUDY

Sustainable Poultry Farming: A Concept of IoT-based Poultry Management System for Small-scale Farmers

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Abstract

Conventional poultry management techniques are failing to meet increased demand for poultry products as the population continues to grow. As a result, this issue has become a major concern for small-scale farmers, particularly those in low-income areas, in terms of food security. One of the main reasons for this is that the farmers rely on intensive farming methods which are inefficient for automating daily poultry operations. However, intensive farming methods pose major environmental concerns to ecosystems and poultry health. Also, the environmental conditions, welfare, and productivity of poultry operations may be harmed by the global climate crisis and poultry waste products can be disastrous to poultry health and need to be managed effectively. This demands for enhanced poultry management systems to alleviate the environmental conditions and boost optimal poultry well-being. This paper presents a compact, low-cost internet of things-based poultry management system for a small –scale farmer. The system employs a synergistic combination of cloud and Internet of Things technologies to monitor and regulate temperature, humidity, water level, ammonia gas concentration, and lighting systems in real time. The hardware prototype was successfully implemented and tested at the Obasanjo poultry farm home in Nigeria. The experimental results demonstrate that the daily temperature is maintained between 27 °C and 32 °C. The relative humidity ranges from 71 % to 72 %, and the ammonia gas (NH₃) level increased intermittently for the first three days before remaining steady, reaching a maximum of 30 ppm. The illumination is optimally adjusted using a pre-configured algorithm to promote maximum egg production and poultry health.

Keywords: Poultry health, Internet of Things, Smart sensors, Lighting strategy, Environmental parameters

1. Introduction

Smart technology has long been a primary focus of sustainable agriculture future [1]. Agriculture sector has substantially enhanced the world's economy growth in terms of food security [2–4], job creation [5–7], raw materials [8–10] and poverty alleviation [11–13]. In addition, this sector accounts for nearly 4 % global GDP, with about 1.4 % attributed to poultry production [14]. The poultry

farming is a major contributor to economy especially in developing countries, where majority rely on livestock production for livelihood and nutritional value. The high demand for poultry products such as eggs and meat in this region is driven by its affordability, nutritional value, and sustainability. Poultry farming in Nigeria holds significant potential for economy growth and food security but faces several challenges that hinders its growth and sustainability. Currently, small-scale farmers rely on

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manual processes for various operations such as feeding, watering, egg collection, health management, disease control, and waste management which are inefficient to enhance productivity and efficiency. Also, the growing population and increasing demand for nutritious food further increases the burden of agricultural systems, which calls for urgent attention for smart agriculture such as smart irrigation system [15], nutrient management [16], smart technology and renewable energy [17], livestock monitoring using agricultural drones and robots [18], and food waste management [19] to enhance food security and preserve environmental integrity. For these reasons, agripreneurs and farmers have begun to focus on strategies to promote poultry sustainability with special emphasis on smart poultry management system and Internet of Things (IoT).

The IoT is an emerging smart technology, which is gaining prominence in wide range of applications including smart farming. The internet of Things is a network of interconnected devices, sensors and other technologies that are used to automate a variety of applications via internet. Similar technologies proposed in the literature for intensive and medium scale poultry farming includes smart sensor [20], 5G cellular network [21,22], radio frequency identification [23], Zig-Bee [24], computer vision and imaging [25,26], blockchain [27], robotic [28] and drones [29,30]. An IoT-based poultry management system seeks to remotely monitor and automate various environmental parameters that directly impact the health and productivity of the birds. Due to the inhospitable climate, it imperious to ensure optimal conditions of poultry house parameters such as temperature, humidity, lighting, air quality and water intake level. Increased temperature levels in poultry houses increase mortality rates and reduce growth performance, eggs production, feed efficiency and survival rates of the chickens. The welfare of layers rely heavily on various ecological variables that may adversely influence the bird egg weight, incubation time and hatching time [31,32]. Unhygienic poultry house may cause accumulation of toxic gasses (NH_3 and CO_2), increase resources competition among the birds and may also increase the chances of spread of infections [33]. The proposed poultry scheme focuses mainly on layers breeding i.e., eggs production for small-scale farmers. United Nations (UN) highlights the need for investing in smart agriculture [34] which includes food shortage mitigation, provision of nutrition for all by 2030 and enhancing of food security. Alongside, the Sustainable Development Goals 2 (SDGs-2) emphasizes the need to

Nomenclature

Ag	Ammonia gas sensor
EPP32SOC	Wireless Enable System on Chip Microcontroller
LED	Light Emitting Diode
Lg	Light sensor
IoT	Internet of Things
SDG	Sustainable Development Goal
Tp	Temperature sensor
HC-SR04	Ultrasonic sensor
UN	United Nation
Wl	Water Level sensor
SonEx	Switching on Extractor
SoffEx	Switching off Extractor
SonFan	Switching on Electric Fan
SoffFan	Switching off Electric Fan
SonHm	Switching on Humidifier
SoffHm	Switching off Humidifier
SonHx	Switching on Heater
SoffHx	Switching off Heater
NH_3	Ammonia gas
CO_2	Carbondioxide
CH_4	Methane
C2-C3	Light hydrocarbon
H_2S	Hydrogen sulfide

end hunger, improve nutrition, and prioritize sustainable agriculture. Also, this study contributes to some the SDGs including access to sufficient and nutritious food, foster individual health and well-being and investment in sustainable agriculture. The motivation of this study lies in meeting the nutritional needs of a growing population while also addressing the challenges of inefficient poultry management caused by conventional practices, as well as exploring smart technology as a solution for profitable and sustainable poultry farming. The research emphasizes on environmental monitoring in poultry farming. This includes boosting egg production with the use of a lighting scheduling system.

The main focus is to ensure sustainable food production system via sustainable agricultural practices with smart technology, which ensure access to nutritious food and improve production of poultry products. This study proposes a cheap and portable IoT-based poultry management system that enables farmers to remotely track, and regulate various environmental parameters that can affect the poultry's health and productivity. Furthermore, the proposed scheme seeks to enhance productivity, reduce losses, and improve overall management of poultry through real-time monitoring for small-scale farmers in developing country like Nigeria. The novel contributions of this paper are summarized as follows.

- The paper utilizes IoT, embedded systems, Bylnk cloud, and advanced algorithms to improve egg production and poultry health.
- In real time, the proposed system supervises and manages vital environmental parameters such as temperature, humidity, light, ammonia gas concentration, and water level.
- The proposed system employs cutting-edge sensors, including the DHT11 humidity and temperature sensor, the HC-SR04 water level sensor, the MQ-2 ammonia gas sensor, the LDR light sensor, and the PH sensor. The system uses the Bylnk cloud to improve user interaction by enabling wireless connectivity between devices and poultry users.
- Small-scale farmers can access real-time environmental data via the internet from anywhere in the world as the sensor data transferred in real-time to the Bylnk cloud and App. This feature enhances health and production while allowing for more accurate poultry monitoring.
- The study incorporates a lighting scheduling approach to optimize egg production.
- The study utilizes smart water level sensors to regulate water levels in storage tanks, resulting in greater water conservation than manual methods.
- The study highlights the significance of smart sensors for poultry health and well-being, as well as their influence on smart farming techniques.

2. Literature reviews

Poultry refers to domesticated birds raised primarily for meat and egg, making it a key component of global food security. Various environmental variables such as temperature, humidity, ammonia gas concentration, lighting, and water supply influence the survival, production (eggs), growth, and management of poultry birds in many ways. Several studies have been conducted on how to improve the poultry welfare as inefficient poultry management results in prolonged health crisis, decreased productivity, economic losses, and high mortality rate of poultry birds [35]. Many poultry environmental variables of interest describe the health and welfare of poultry but only a few variables are of significant of interest.

2.1. Temperature

In poultry, temperature refers to the thermal environment in which poultry is raised. Temperature control is a vital aspect of poultry management

as it directly affects the health, growth, and productivity of poultry. Birds are warm-blooded species, and they assume uniform internal body temperature [36]. Physiological processes such as respiration, feeding, metabolism, egg production, growth and general behaviour are affected by the temperature [37]. Maintaining optimal temperature level is essential for the well-being of poultry, for instance, broiler chicks typically require a temperature of about 32–35 °C during the first week of life, gradually decreasing as they grow. Layers, generally thrive in a temperature between 18 and 24 °C [38–40]. However, poultry birds are sensitive to temperature changes for instance, extreme temperature can significantly affect egg production, shell quality, and egg weight in layers, and low temperature in layers distorts their metabolism process leading to nutritional deficiencies.

2.2. Humidity

Humidity refers to the amount of moisture present in the air within poultry environment. It is a vital environmental parameter that affects poultry welfare. The major source of humidity in poultry facilities is typically poultry manure, which is a composition of Ammonia, NH_3 , Carbon dioxide, CO_2 , methane, CH_4 , light hydrocarbon, $\text{C}_2\text{--C}_3$ and Hydrogen sulfide, H_2S [41]. Other sources includes litter moisture content, air quality, ventilation, and gas concentration. Relative humidity is used to measure humidity level, indicating the amount of moisture in the air relative to the maximum amount of moisture the air can hold at a given temperature. The acceptable humidity level for optimum poultry condition is within the range of 50 %–70 % [42]. Also, the impact of humidity on poultry has different consequences depending on humidity levels, humidity within the optimal range helps to minimize respiratory diseases, increase growth, and enhance cooling mechanisms and lower dehydration. However, high humidity level can breed ground for harmful microorganisms, increased cold stress and often lead to decrease egg quality due to increase stress hormones in eggs [43].

2.3. Lighting

Lighting refers to the use of artificial or natural light to create an optimal environment for poultry growth, health, and productivity. In poultry, proper lighting is a major concern as it affects poultry's behaviour, reproduction, egg shell quality and welfare [44]. It plays a crucial role in regulating the biological rhythms of poultry and also influences

behaviour such as feeding, mating and egg-laying. Lighting significantly affects egg production in layer i.e., birds exposure to increase daylight tend to produce more eggs. However, insufficient light may lead to stress and health issues while excessive brightness can result in discomfort [45]. Various types of lighting sources such as incandescent, fluorescent, and light emitting diode (LED) light serve different purposes [44] as presented in Table 1. LED is a preferred choice in poultry facilities due to its energy efficiency and light quality. The effect of varying light intensity in poultry health and growth is shown in Table 2.

2.4. Ammonia gas concentration

Ammonia is a significant by-product in the decomposition of nitrogenous compound such as uric, urea nitrogen, and protein in poultry manure [52]. It is produced when organic matter in poultry litter such as manure and bedding materials, breakdown. The amount of ammonia excreted by poultry is directly proportional to the protein content of their diet. The decomposition of other organic matter such as uric acid, feed spills, dead

birds, organic debris also contribute to ammonia accumulation in the poultry. The unused protein level in poultry diet is excreted in feces and later degraded, releasing NH_4^+ , which is further converted to NH_3 .



When ammonia accumulates to toxic level, the NH_3 gas reacts with the existing humidity to form corrosive NH_4^+ solution, which affect the respiratory tract of the birds. However, if the ammonia concentration is not controlled, the poultry can ultimately cause inflammation in the lungs or air sac, leading to severe death [53]. Even at survivable concentrations, excess ammonia can lead to respiratory problem, impaired immune function, reduce growth and egg production [54,55]. In poultry, the harmless limit for NH_3 is 25 ppm and the litter PH is as metric to measure ammonia concentration of poultry facilities. The ammonia concentration and litter PH with their effect are presented in Tables 3 and 4.

2.5. Water supply

Water, chemically represented as H_2O is an essential nutrient in poultry farming. It is crucial in various physiological and metabolic processes in

Table 1. Different lighting sources for poultry management system.

Color	Effects	Species
Red light	For sexual maturity and egg production	Layers
Blue light	Stress reduction and to improve broiler welfare	Broiler
UV lights	Influence chickens behaviour and well-being	Layer and broiler
Fluorescent light	Enhance egg production and maintain stable environment	Layer
LED light	Enhance production performance and egg quality Improve broiler chicken welfare, feed conversion rate and productivity	Layer Broiler
CLF light	Influence poultry behaviour, growth rate and reproductive performance	Layer
Combination of light color	Improve growth and productive performance in broiler	Broiler

Table 2. Poultry light intensity on poultry species.

Species	Duration (hrs./day)	Light intensity (Lux)	References
Broiler	16	10–20	[46,47]
Layers	14–16	10–30	[48,49]
Growing pullets	24 (first week), 9 (after six weeks)	10–15, 30–50	[50,51]

Table 3. Ammonia concentration in poultry house and their effect.

Ammonia concentration	Ammonia level	Effect
<10 ppm	Optimal	Best optimal condition for good growth
<10–25 ppm	Threshold	Optimal condition for bird's growth
>25 ppm	High	Condition that leads to significant health issues such as respiratory distress, decreased growth and increased susceptibility to infections and diseases.
>50 ppm	Critical	Risk of severe respiratory damage and other health complications increase dramatically.

Table 4. Ammonia litter PH level and their effect.

NH_3 PH level	Effect
6.0–7.0	Optimal growth for poultry birds
6.8	Ammonia begins to volatilize significantly
7.0–8.5	Increase ammonia emission
>8.5	Respiratory issues and other health concerns for the birds.

poultry. Water is vital for thermoregulation, waste elimination, and also serves as a vehicle for nutritional additives [56]. Birds require a constant supply of fresh, clean water to maintain hydration and health. Water aids in softening feed in crop, facilitating its transit through digestive tract, and ensuring the efficient nutrient uptake. The quality and availability of water directly affect the health and productivity of poultry [57]. In addition, clean and uncontaminated water aids optimal growth rate, egg production and bird reproductive performance [58]. The chemical properties of water, including its PH, mineral content, and the presence of contaminants play a vital role in poultry management. Contaminated water can lead to disease outbreaks, while water with excessive minerals can cause scaling in water system.

2.6. Related works

Numerous researchers have adopted internet of Things in poultry farming. Khairul et al. [59] developed an IoT system for poultry farming. The proposed system used smart sensors to detect temperature, humidity, water level and food and also incorporated an alarm system which notify the farmers on the state of the poultry. However, the system can be enhanced with gas sensors to monitor the toxic gas level in the poultry house. [60], used sensors to determine PH value, yolk index, specific gravity of eggs and nitrogen content. IoT has been reported in literatures to detect poultry diseases [61] and monitor air quality [62], light intensity [63], CO₂ concentration [64], poultry welfare [65], water level [66], chicken behaviour [67], ammonia gas concentration [68], egg production [69,70], chicken cage [71], food quality control [72] and PH level [73]. Chigwada et al. [74] presented an IoT-based poultry system for small-medium scale producer. Their work used various smart sensor to monitor vital environmental conditions that affect poultry health and welfare. The system features a novel light scheduling and automatic switching control. An electric power control strategies for minimizing poultry farming is presented in Ref. [72]. Their proposed strategy utilized a novel algorithm to minimize the power consumption and distributed water supply, and poultry farm's environment. The use of smart sensors [75–77], bid data [78], machine learning [79–81], artificial intelligence [82–84], edge computing [1,85], deep learning [86], and hybrid AI and IoT [87] is currently gaining significant interest in poultry farming. Manshor et al. [88], propose a Raspberry Pi poultry monitoring system. The system not only monitors the temperature and

humidity levels of the poultry house but also minimize energy consumption within the poultry. It is equipped with alarm system which notify the farmers on the potential danger of environmental parameters variation and excess power consumption. [89], presented a poultry management scheme that uses IoT sensors to monitor poultry's house environmental conditions such as temperature, humidity, wind speed, and CO₂ emission and minimize energy usage. The scheme uses a programmable logic controller ECM6L45160 as micro-controller to automate the operation of exhaust fans and establish a connection sensors and cloud server. The system is further embedded with an adjustable power control mechanism and gas sensor that minimizes the power consumption and toxic gas level within the poultry facility. Elmoulat et al. [90], presented an hybrid algorithms of edge computing and Artificial intelligence to monitor and predict poultry's air quality. Their work focuses on sensing different gas concentrations with various sensors such as CH₄ (MQ-4), NH₃ (MQ-137), CO (MQ-7), and CO₂ (MG-811). It also features a nutritional sensor such as water level and load cell sensors that monitor the chicken water and feed levels. The various works cited in Table 5 presents the concepts related to Internet of Things and framework in poultry farming.

3. Materials and method

The internet of things based poultry management system (IoT-PMs) is an efficient and budget-friendly cloud enabled scheme that monitor and regulate various poultry environmental variables such as temperature, humidity, water level, and lighting. The IoT-PMs composed of environmental sensors, actuators, wireless enable system on chip micro-controller (EPS 32 SOC), and internet connectivity and Bylnk sever as shown in Fig. 1. Also, the system is equipped with biosecurity, lithium-ion battery, and smart alarm notification system. The use of bylnk IoT platform access via a smartphone facilitated the real-time monitoring and ease of accessibility of the proposed system. The unique features of lighting scheduling and real-time control of poultry ecological parameters enhance the sustainability of poultry welfare and egg production. The IoT-PMs hardware and software components are highlighted as follows:

3.1. System hardware design

As shown in Fig. 1, the proposed IoT-PMs consists of various sensors which have been integrated with

Table 5. Previous related studies on poultry management systems.

S/N	Highlights	Remarks	Citation
1	The study leverages on Artificial Intelligence (AI), edge computing and IoT for prediction of toxic gas concentration in poultry barn	Lacks real-time monitoring of other vital environmental parameters such as temperature, humidity, water level, and lighting.	[91]
2	Proposed an advanced algorithm and IoT system for real-time monitoring of humidity and temperature of poultry farm	Lacks environmental sensors for ammonia gas level monitoring and lacks implementation of remote controller and alert notification system.	[92]
3	Proposed a solution of smart sensor and GSM technology for intelligent poultry system. It also uses GPRS to provide seamless connectivity to the cloud server	Lacks practical integration of the proposed model in real-life setting. Also, farmers lack remote access to vital environmental parameters since sensors for these parameters are not considered.	[93]
4	Proposed integrated solution of wireless sensor, mobile application, and cloud computing for poultry health. Also, it focuses on four sensing parameters (1) temperature, (2) humidity, (3) light intensity, (4) population density.	The model relies on logic controller and cellular network of 3G and 4G. Also, lacks practical application of industry 4.0 and IoT platform for real-time visualization and analysis.	[94]
5	Proposed a cloud-based system for poultry farming while highlighting the significant benefit of cloud and wireless technologies such as (1) wireless sensor network, (2) sensors, (3) MySQL 5.7, (4) Wi-Fi, and (5) IoT devices.	Lacks real-time monitoring of environmental conditions due to the absence of industry 4.0 and IoT platform for remote data visualization.	[95]
6	Proposed IoT and sensory system for poultry health monitoring. It also, uses mobile ThingSpeak application for precise data visualization and analytics.	Lacks security measures such as alert/warning notification system to trigger farmers' awareness for potential dangers.	[96]
7	Proposed smart sensing technology for efficient poultry management while leveraging on Internet of Things for data sharing, visualization and analysis.	There is no consideration for practical uses. The use of wireless sensor technology raises the danger of radio frequency interference, which could disrupt the data stream.	[97]
8	Proposed an AI-driven automated poultry monitoring system for precise chicken counting. It also uses AI algorithms that enhance resource allocation and operational efficiency.	Scalability issues, latency issues due to real-time detection of large data set. Also, practical usability of the system can be compromised.	[98]
9	Proposed an integrated technology-driven solution of embedded framework and smartphone via internet for poultry health monitoring. The study also, highlights smart distribution and environmental monitoring of poultry farming.	Lacks real-time graphical display of data for visualization and analysis.	[99]
10	Proposed IoT, wireless sensor networks and Artificial Intelligence system for enhancing food sustainability and maintaining environmental integrity.	Lacks lighting scheduling strategy	[100]
11	Proposed GSM-based system for efficient poultry while emphasizing on the benefit of (1) GSM, (2) smart sensors, (3) energy storage system	Lacks industry 4.0's real application and visual, graphical, and numerical display of real-time poultry data for precise monitoring and control.	[101]
12	Proposed an integrated system for poultry health monitoring while exploring innovative technologies such as (1) smart sensors (2) internet of things, (3) bio sensors	Sensor and IoT device interoperability concerns, as well as scalability	[75]

the EPS32 microcontroller, Wi-Fi connectivity, and programmed to provide an economic, reliable sustainable means of monitoring and controlling poultry birds remotely. The e-MHs hardware and software components are highlighted as follows:

3.2. IoT input sensors

A “DHT11 sensor” was deployed to provide accurate readings of both temperature and humidity levels within the poultry facility. A fan is

automatically activated to compensate for real-time poultry temperature above 18–24 °C. A relative humidity above 70 % will reduce hens' reproductive performance and may lead to thermal discomfort and chicks behaviour changes. On contrary, a low relative humidity can reduce egg size and quality and also affect egg's moisture content, resulting in cracking problems. In case of extreme humidity level, an extractor serves effective moisture expellant, regulating temperature and improve air quality.

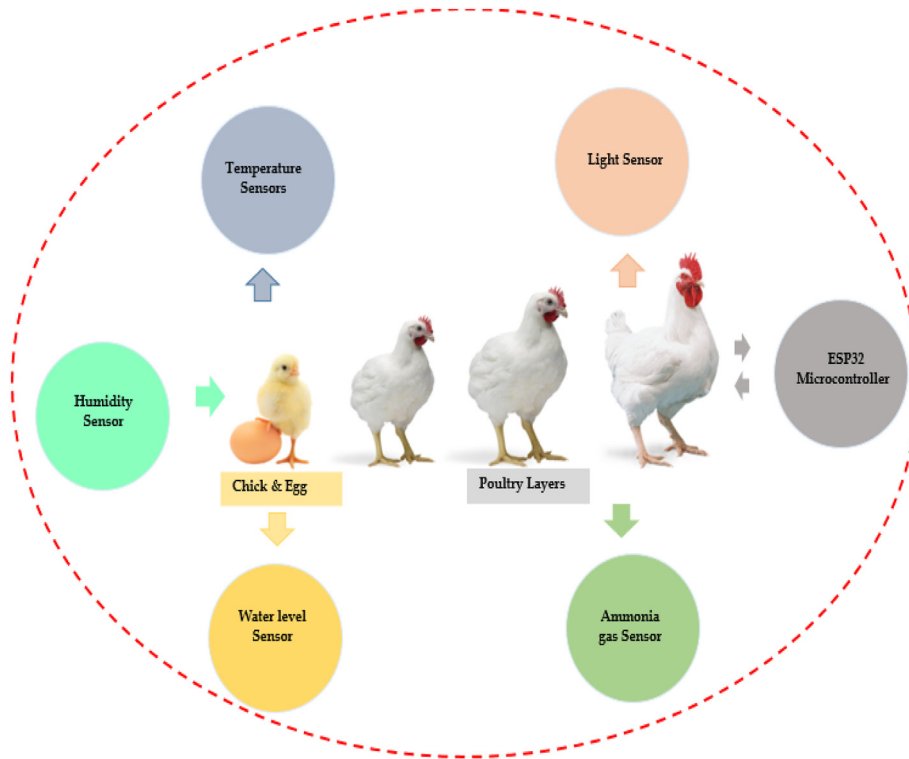


Fig. 1. Hardware sensors integrated with EPS32 SOC Microcontroller.

The “MQ-2 gas sensor” was used to detect and track the poultry’s facilities ammonia (NH_3) gas concentration level. Once the ammonia gas level exceeds the threshold limit of 25 ppm, the ESP32 SOC microcontroller automatically energizes the relay to switch ON the extractor fan which removes excess ammonia gas from the poultry facilities.

3.3. Light scheduling and power control

Light scheduling plays a crucial role in influencing egg production in poultry layers. Optimal light scheduling can enhance egg production, immune response, feed conversion rate and birds well-being. The proposed scheme has the capability to schedule light timing activities and save energy and costs. The system is programmed to detect the light intensity of the poultry house and regulates it based on the intensity density. The system uses a light dependent resistor (LDR) to monitor and regulate the lighting conditions of poultry layers via a programmable microcontroller. LDR resistance varies in response to light intensity.

3.4. Water supply control

An ultrasonic sensor was utilized to detect water level below 40 cm from the sensor. The system uses

a pre-defined commands to activate the pump whenever the water level in the water feed tank goes below the minimum set-point. In addition, the microcontroller is programmed to deactivate the pump when the water level reaches the maximum set-point.

3.5. Microcontroller

The “ESP32 SOC” served as a microcontroller for this study and was a preferred candidate due to its Wi-Fi and dual-mode Bluetooth capabilities. The microcontroller provides a wide range of functions and capabilities, making it well-suited for IoT-based poultry management systems. The “ESP32” features a dual-core Tensilica Xtensa LX6, CPU, 520 KB of volatile memory, and 802.11 b/g/n Wi-Fi.

3.6. Voltage conversion module and battery

The electronic chip “LM2596” controls voltage in light-emitting diodes, batteries, power supplies, and other devices. The IoT-PM is powered by two 3.5 V 18,650 lithium batteries connected in series.

3.7. Output devices

The poultry scheme consists of different output devices such as pump, extractor fan, fan, buzzer,

and lights. The ESP32-microcontroller coordinate and control the switching processes of the output devices based on the pre-defined threshold levels.

3.8. User devices

Smart electronic devices such as smartphones, iPads, computers, notepads, and tablets allow farmers/producers to access, interact with, and monitor environmental conditions and water levels on the bylnk platform in real time over the internet. Fig. 2 depicts the system's hardware components.

3.9. Cloud server

The study employs the freely available Blynk cloud service for remote monitoring and control of IoT-powered devices. To utilize the Blynk app (<https://play.google.com/store/apps/details?id=cloudblynk> domain), first set up your network name and password.

3.10. System hardware design

The Arduino platform was used to generate C++ programming code for the ESP32 microcontroller. All other actuators and input sensors were correctly coded. The platform aims to simplify code and make it less boring. The open-source Arduino software platform consists of two main components: the Integrated Development Environment (IDE) and a core library. The IDE contains a simple editor with a number of tools and features for programming Arduino boards. Programming functions are conveniently accessible via the toolbar.

3.11. Blynk platform

The Blynk platform facilitates real-time monitoring and control of poultry facility from anywhere via internet. Blynk is a powerful low-code platform designed to control and manage connected IoT-PMs devices. Blynk App had to be installed on the smartphones or laptops via Android emulator to establish an interactive environment with the

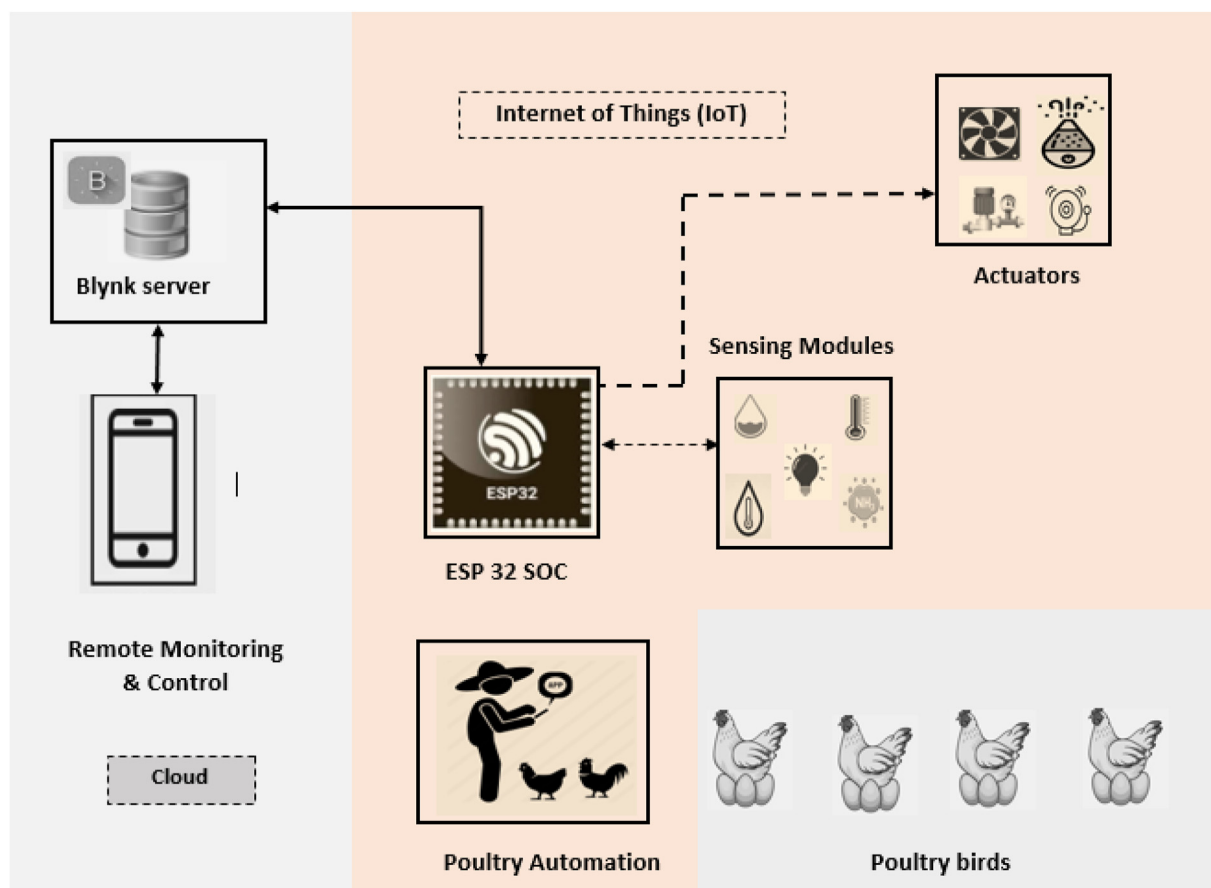


Fig. 2. Proposed system architecture.

dashboard. This App can be installed with mobile/desktop operating systems like iOS/Android, Windows, MacOS, and Linux. It is also compatible with various embedded systems such as Raspberry Pi, Zephyr, particle photon, Zig-Bee and ARN Mbed. The Android mobile OS is selected in this study because it permit farmers/producers easy access to monitor, interact, manage, and control environmental parameters that affect the welfare and productivity of the poultry remotely via smartphones.

3.12. The proposed model concept

The proposed model uses four-tier Internet of Things operational layers (A, B, C, and D). The tier A is the sensing layer (sensing modules), which provides vital data related to the poultry facilities environmental parameters such as temperature, humidity, lighting, water level and ammonia gas level. The tier B is the middleware layer, which establishes a data-sharing link between sensing modules data and the EPS 32 SOC microcontroller as shown in Fig. 2.

The tier C is the network layer that uses wireless technology to establish remote communication between the sensing modules sensors and actuators from system –on –chip microcontroller to cloud database server. The tier D is the application layer, which is responsible for Wi-Fi-enabled IoT cloud platform and to provide farmers with real-time access to graphical charts and instants numeric value of poultry parameters, alert, and updates. In this study, the Blynk server is hosted on the ESP 32 SOC

microcontroller, which also serves as cloud storage accessible via a Wi-Fi connectivity. The visualization of the proposed model requires two main steps, remote connection of the farmer to IoT platform via Blynk app and data storage on IoT cloud server. The Blynk stores and read collected data from the database and display the corresponding time series graphs, and instants values as shown in Fig. 3.

3.13. Circuit design and simulation

The experimental model begins with the Arduino frizzing software circuit design and simulation. The circuit schematic for the poultry management system is depicted in Fig. 4. The input sensors are interlinked with the EPS32 microcontroller via internet. The EPS32 controller provides a real-time data sharing and transmission of various input sensor readings to the blynk server using Wi-Fi connectivity. Also the system makes use of electric-powered fans and extractors to compensate for the poultry facility's severe temperatures and humidity. The system makes use of electric-powered fans and extractors to compensate for the poultry facility's severe temperatures and humidity.

3.14. Coding and Blynk configuration

The second step is to program the EPS32 microcontroller and input sensors, after which the compiled files are transferred to the EPS32 SOC microcontroller via the Arduino IDE 2. The control panel runs on the Blynk server. The Blynk app was

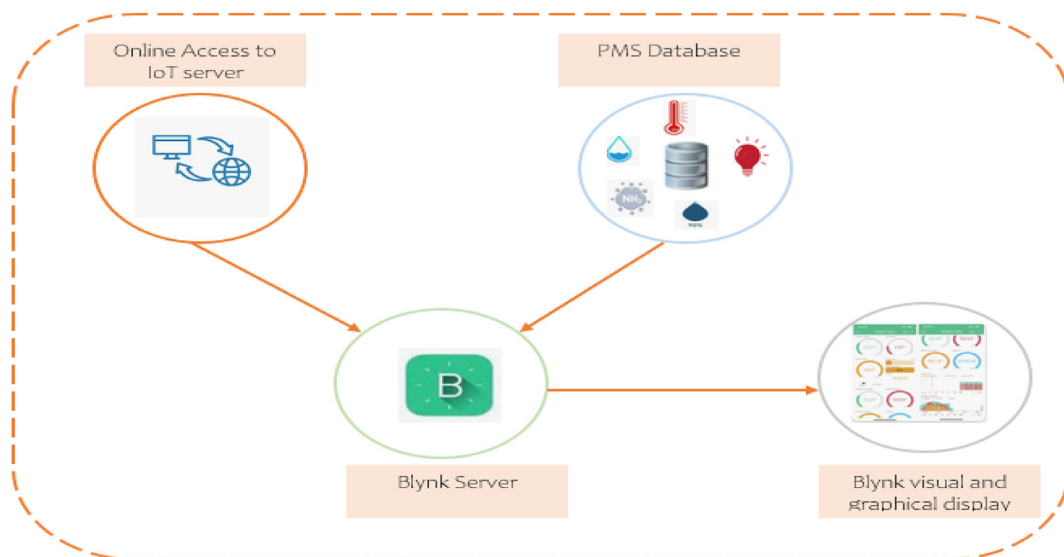


Fig. 3. Steps in proposed model visualization.

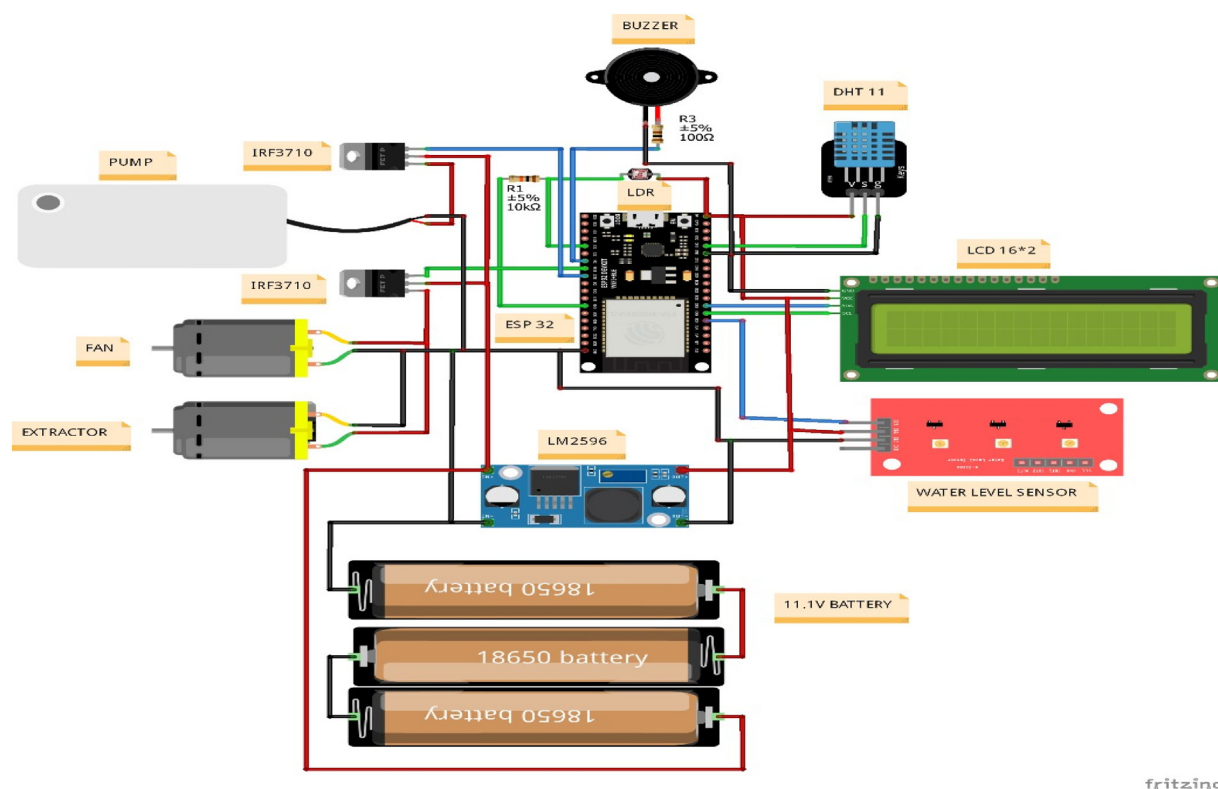


Fig. 4. Proposed system hardware schematic diagram.

installed on the smartphone, and EPS32 was linked to the app's local server via Wi-Fi and a Blynk authentication token. In addition, the "Liquid Crystal" file was uploaded and initialized, and the LCD was connected to the microcontroller via pins.

3.15. Pseudo-code

The pseudo-code in Table 6, explains the functionality of the poultry scheme; it defines the simulation of the smart poultry management system based on an embedded system that monitors environmental conditions, controls pumps, humidifiers, fans, and communication with the cloud (Blynk platform and application). The pseudo-code begins by declaring the minimum threshold for several sensors and then uses variables to indicate sensor reading, pump state, fan state, heater state, and extractor state, as shown in lines 1–18. Lines 18–20 configure the embedded system on chip (ESP32 SOC) and sensors. In lines 21–72, the while loop is executed repeatedly to simulate the proposed system's real-time behaviour. Sensor data is collected continually within the while loop, employing unique sensing algorithms for temperature, humidity, ammonia gas, light, and water level. The

collected data is formatted and sent in real time to the Blynk app and dashboard, allowing for display and analytics. Lines 27–30 create wireless connectivity between the ESP32 SOC and the cloud server (Blynk Cloud). The lines use special algorithms embedded in the ESP32 SOC establish a secure connection to the cloud. Line 31–54 test the reliability of the pump control, fan control, heater control, and extractor control (actuators) for simulated water level, humidity level, and temperature and ammonia gas sensors. The actuator control flags (SonTp, SoffTp, SonFan, SoffFan, SonHt, SoffHt, SonEx, SoffEx) are configured and simulated using the actuators' switching state (ON/OFF). Lines 55–69 define pre-configured lighting algorithms for poultry layers dependent on their age (week). A 5 s delay is added for loop iterations.

3.16. Integration of the propose poultry model in live poultry farm

The proposed scheme was integrated into the existing poultry facilities of Obasanjo poultry farm, covering an area of 9000 square feet (300 × 30). The farm focusses mostly on the battery cage technique for egg production. The farm infrastructure includes

Table 6. Algorithm.

Pseudo-code of the Proposed Poultry Scheme

```

// Define Constants:
1: Min_Temperature_Threshold = 30
2: Min_Humidity_Threshold = 70
3: Min_Ammonia_Gas_Threshold = 25
4: Min_Water_Level_threshold = 6
// Variables:
5: Tp
6: Wl
7: Ag
8: Lg
9: SonTp = false
10: SoffTp = false
11: SonHm = false
12: SoffHm = false
13: SonFan = false
14: SoffFan = false
15: SonHt = false
16: SoffHt = false
17: SonEx = false
18: SoffEx = false
17: Max = 50
18: Initialize ESP32 SOC
19: Initialize sensors
20: While true
//Simulate sensor readings:
21: Tp = GetTemperature()
22: Hm = GetHumidity()
23: Ag = GetAmmoniaGas()
24: Wl = GetWaterLevel()
25: Lg = Getlight()
//Simulate Bylnk App and Bylnk web Communication
26: if (ESP32 SOC is connected to Bylnk IoT cloud)
27: print collected data on the laptop
28: send collected data to Bylnk platform
29: else:
30: Try to re-connect
//Control of tank pump based on water level
31: if (WaterLevel  $\leq$  Min_Water_Level_Threshold)
32: SonTp = true
33: SoffTp = false
34: else:
35: SonTp = false
36: SoffTp = true
//Control of electric fan based on temperature
37: if (Temperature  $\geq$  Min_Temperature_Threshold)
38: SonFan = true
39: SoffFan = false
40: else:
41: SonFan = false
42: SoffFan = true
//Control of electric heater based on temperature
43: if (Temperature  $\leq$  Min_Temperature_threshold)
44: SonHt = true
45: SoffHt = false
46: else:
47: SonHt = false
48: SoffHt = true
//Control of Ammonia gas based on relay
49: if (AmmoniaGas  $\geq$  Min_Ammonia_Gas_Threshold)
50: SonEx = true
51: SoffEx = false
52: else:

```

```

//Temperature
//Water level
//Ammonia gas
//Light
//Tank pump ON state
//Tank pump OFF state
//Humidifier ON state
//Humidifier OFF state
//Fan ON state
//Fan OFF state
//Electric Heater ON state
//Electric Heater OFF state
//Extractor fan ON state
//Extractor fan OFF state
//Maximum distance from the sensor
//System on chip Wi-Fi module
//initialization of (Tp, Hm, Ag, Lg, Wl)

//read temperature value (Tp)
//read humidity value (Hm)
//read ammonia gas value (Ag)
//read water level value (Wl)
//read light level value (Lg)

```

(continued on next page)

Table 6. (continued)

Pseudo-code of the Proposed Poultry Scheme

```

53: SonEx = false
54: SoffEx = true
//Control of lighting based on the age of the layers
55: if (The age of the Layer  $\leq$  18 weeks)
56: Switch ON the light for a period of 19.00–20.00
57: else:
58: use the artificial light
59: if (The age of the Layer  $\leq$  22 weeks)
60: Switch ON the light for a period of 19.00–21.00
61: else:
62: use the artificial light
63: if (The age of the Layer  $\leq$  32 weeks)
64: Switch ON the light for a period of 19.00–22.00
65: else:
66: use the artificial light
67: if (The age of the Layer  $\leq$  42 weeks)
68: Switch ON the light for a period of 19.00–20.00
69: else:
70: use the artificial light
71: Delay (5s)
72: End while

```

//repeat for each period for 5s

three rows of chicken laying egg cage systems with a 1-m gap for biosecurity precautions. The complete implementation of the poultry scheme includes sensors for temperature, humidity, water level, ammonia gas and real-time IoT cloud based server. Also, 24 h horizon for a period of 1 week was chosen as the testing time. The scheme was installed in the poultry facilities at a distance of 1.5 m on the floor and connected to the cloud server via the internet. This allows farmers to remotely obtain environmental information about poultry farms. The on-site environment of the poultry facilities is presented in Fig. 5. Furthermore, the Blynk server is accessible to farmers via smart devices like smartphones, iPads

and tablets and password protected to prevent unauthorized user access to valuable data. Only authorized user to establish an encrypted connection to the cloud server.

In addition, the system helps farmers or agripreneurs to monitor real-time environmental data and the poultry birds' health condition via mobile application. With these information, producers can determine the optimal condition of poultry facilities and make informed decisions about the health of their birds. However, there is an increasing demand for precision agriculture. Recently, significant attention has been focused on sustainable poultry farming solutions [102], such as Radio-frequency



Fig. 5. Live implementation of the proposed system.

identification (RFID) [103], smart sensors for optimal health and livestock well-being have been proposed in Refs. [28,69,84,92,104–107]. Researchers are actively focusing on livestock behaviour monitoring [32,67], Nanosensors for disease detection [108,109], wearable sensors for livestock identification, tracking and health monitoring [20,110], and nanotechnology [111,112]. Future research also tends towards developing data prediction techniques using emerging technology like Artificial intelligence (AI), machine learning (ML), deep learning and Edge computing [79,91,113] to make smart decision on poultry health and welfare based on real-time data and trends.

4. Results and discussions

The Blynk dashboard has been customized to provide a quick overview of poultry management system data value in numerical and graphical formats. Numerical indicators in the form of gauges, as shown in Fig. 6, provide for easy observation of real-time data on temperature, humidity, illumination, ammonia gas level, and water level.

4.1. Temperature and humidity control

In poultry facilities, a “DHT1” is used to measure humidity and temperature. For a week, the temperature and humidity of the poultry facilities are monitored. Each sensor is sampled every 15 min for one week, with a total of 96 sampling intervals (N per 24-h horizon). Temperature and humidity are critical environmental variables that must be continuously monitored. In fact, operating the system above the safe temperature and humidity levels can have a direct impact on the bird's optimal health, egg production, shell quality, and metabolic

process. When the temperature reaches the safe limit of 30 °C, as shown in Fig. 7, a fan is activated to cool the poultry facilities and keep the temperature within acceptable limits. Additionally, the electric heater is engaged when the temperature falls below 12 °C, compensating for heat loss. In addition, the system ensures that the humidity is within the acceptable threshold. If the humidity level exceeds 70 %, as indicated in Fig. 7, the extractor fan is activated to remove surplus air. The snapshot in Figs. 6–7 show that the poultry facilities temperature varies from 29.5 °C to 31.8 °C for a period of 168 (24 × 7) hours. The experimental daily average readings of poultry facility temperature and humidity are presented in Tables 7 and 8.

4.2. MQ-2 gas sensor

The experimental set contained a “MQ-2” gas sensor that detected and tracked the levels of ammonia (NH₃) in the poultry farmhouse. When the ammonia gas level above the 25 pmm threshold limit, the ESP32 SOC microcontroller activates the relay to turn on the extractor fan, eliminating excess ammonia gas from the poultry facility. Furthermore, the buzzer is activated to produce a sound indication and bring farmers' immediate attention to the prospective threats of stored NH₃ gas. When the ammonia gas levels in the poultry facilities reach the recommended (safe) level, the buzzer and extractor fan are turned off. Table 9 shows the daily average ammonia gas readings from the poultry farmhouse.

4.3. Water consumption monitoring and control

Water is one of the main nutrients needed in macro quantity for poultry bird survival. It is an essential ingredient for thermoregulation and waste

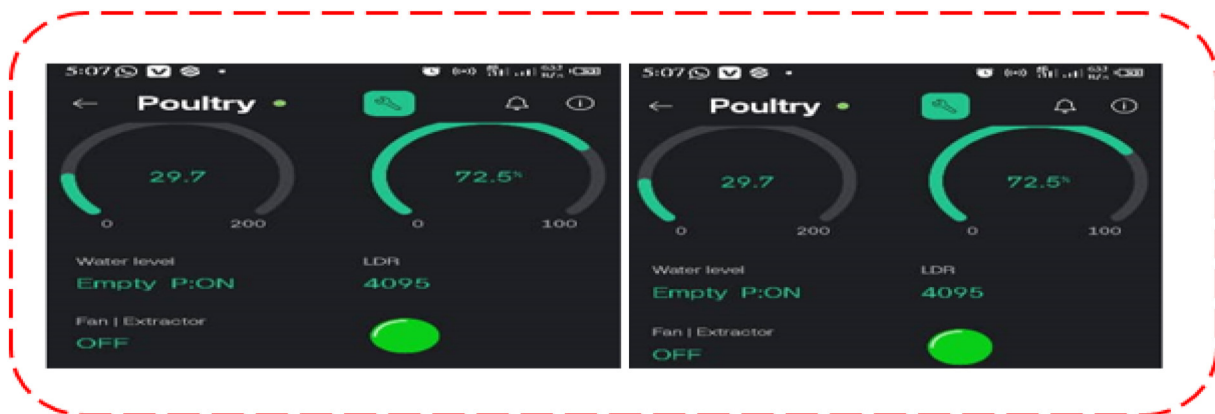


Fig. 6. Snapshot of real-time data in the bylnk dashboard.



Fig. 7. Snapshot of graphical and numerical displays of temperature and relative humidity.

Table 7. Poultry facilities temperature readings (experimental).

Sample (in day)	DHT11 (temperature) Date	Degree Celsius (°C)
Day 1	2024:06–30	27.4
Day 2	2024:07–01	31.00
Day 3	2024:07–02	31.02
Day 4	2024:07–03	31.06
Day 5	2024:07–04	31.08
Day 6	2024:07–05	32.00
Day 7	2024:07–06	32.04

Table 8. Poultry facilities humidity readings (experimental).

Sample (in day)	DHT11 (humidity) Date	g/kg (%)
Day 1	2024:06–30	71.06
Day 2	2024:07–01	71.60
Day 3	2024:07–02	71.80
Day 4	2024:07–03	71.96
Day 5	2024:07–04	72.10
Day 6	2024:07–05	72.30
Day 7	2024:07–06	72.50

removal in the metabolic process. Throughout the week, the poultry facilities' water usage was continuously monitored over a 24-h period. The water level in the poultry storage tank is monitored using HC-SR04 ultrasonic sensors. The minimal restriction of 6 % implies “empty tank,” and the

optimal limit of 95 % represents “full capacity.” For example, the water pump indicated in Fig. 8 is triggered because the water level is less than the minimal threshold of 6 %. As a result, the ESP 32 SOC microcontroller operates the water pump based on ultrasonic sensor data, filling the water

Table 9. Poultry facilities Ammonia and PH levels.

Sample (in day)	MQ-2 Date	PH level (Scale 0–13)	Ammonia gas level (ppm)
Day 1	2024:06–30	6.0	21
Day 2	2024:07–01	6.0	23
Day 3	2024:07–02	6.0	24
Day 4	2024:07–03	7.0	30
Day 5	2024:07–04	7.0	30
Day 6	2024:07–05	7.0	30
Day 7	2024:07–06	7.0	30

storage tank to capacity. When the water storage tank reaches its full capacity, it will automatically shut off. Fig. 9 depicts real-time temperature and humidity data from the experimental setup.

4.4. Alert notification

Aside from the simple visualization of environmental variables (temperature, humidity, light, ammonia gas concentration, and water level). An alert notification system has been developed to generate warning messages when environmental parameters fall outside of the threshold range. Farmers may now not only respond to significant changes, but also make informed judgements about the health and well-being of poultry birds.

4.5. Light scheduling strategy

The proposed technique is integrated with a lighting scheme to achieve optimal egg production. The ESP32 SOC is pre-configured with a light scheduling strategy that follows Table 10. The

experimental approach employs four distinct layer selectors—A, B, C, and D—to represent the chosen week of age for layers 18, 22, 32, and 42, respectively. In addition, the light is automatically turned on and off based on the preconfigured light duration of the selected layers, and the cycle repeats daily until the farmer resets it for new bird selection. When layers are exposed to enough light for at least 15 h per day, they become extremely productive. The sun offers adequate lighting during the day. The lighting scheduling system makes up for low light levels in the evening. Fig. 10 shows the pictorial view of the proposed lighting strategy and Fig. 11 presents graphical representations of the experimental findings of input sensors.

4.6. Performance assessment

Table 11 shows the performance assessment of the proposed system and the existing models. This work proposes an intelligent light scheduling technique for optimum poultry illumination conditions. This work combines IoT, cloud computing. The synergistic integration of these technologies allows for real-time monitoring, remote access, visualization, and analysis of environmental data via the Blynk dashboard and mobile application.

4.7. Strengths of the proposed system

The main strength of the proposed system lies on the use of hybrid technologies of Internet of Things, IoT-cloud, and advanced algorithms to provide real-

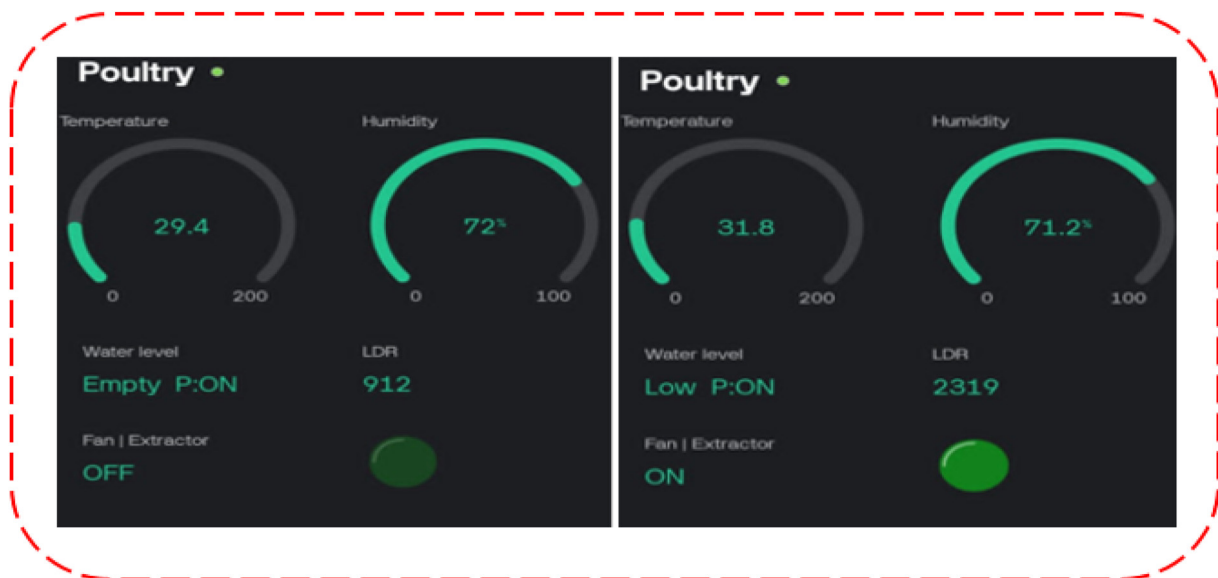


Fig. 8. Snapshot of water level status.

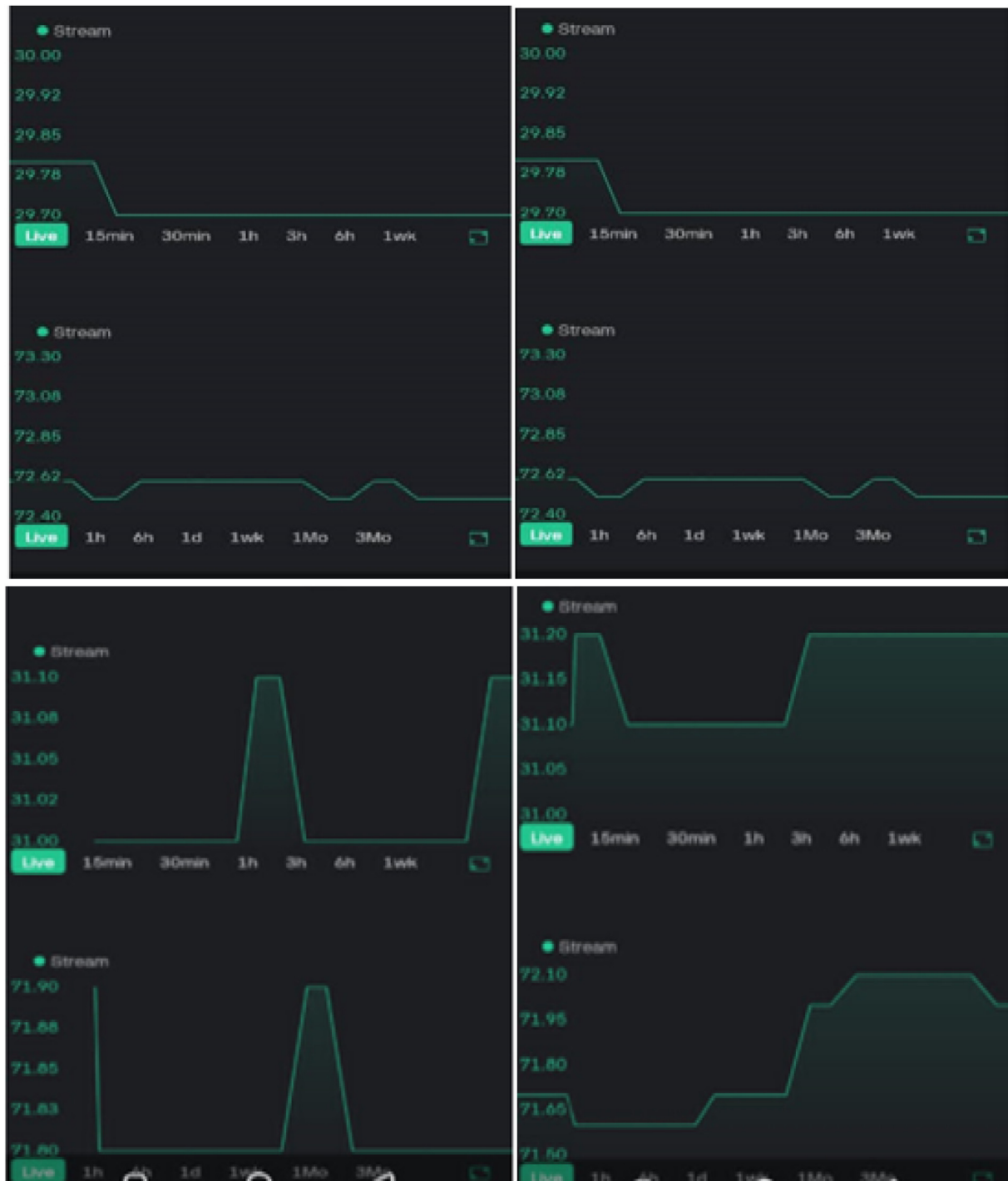


Fig. 9. Graphically display of real-time temperature and humidity data.

Table 10. Light strategy for egg production in layer species.

Layer selector	Age (week)	Light duration of Artificial lighting (hours)	Lighting strategy Peak –time for lighting (hours)
A	18	2	19.00–20.00
B	22	6.0	19.00–21.00
C	32	6.0	19.00–22.00
D	42	7.0	19.00–23.00

time monitoring, remote access and instant alert notification on environmental conditions of poultry farm and health. In addition, the system redefines poultry farming through real time monitoring, rapid response, and remote accessibility, resulting in enhanced productivity and food sustainability. With the Blynk dashboard, users can monitor environmental data that impact poultry's welfare through

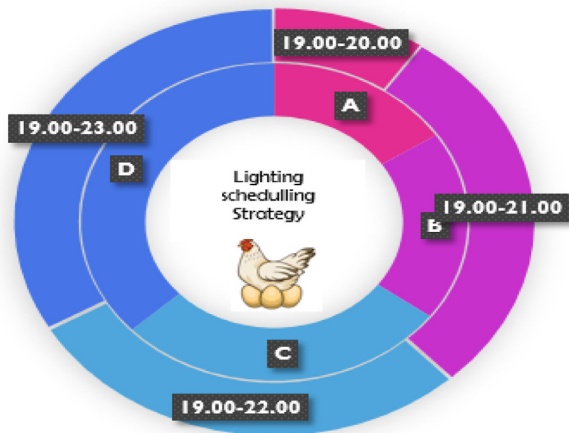


Fig. 10. Pictorial view of the proposed lighting strategy.

mobile application remotely. This eliminates the need for manual checks and provides convenience. Furthermore, the proposed system can detect abnormality in environmental factor or water level and send immediate alerts notification to the users on real-time. The system incorporated lighting scheduling mechanisms to provide an additional layer of

safety for layers breeding that might affect their poultry's behaviour, reproduction, egg shell quality, and welfare. Contrary to the previous works, the proposed system seeks to enhance food security and sustainability using a fused internet of things and smart sensors and advanced lighting scheduling algorithms. A novel alert notification, web-based/mobile dashboard mechanism, and IoT cloud platform incorporated to the system to enhance data update and data sustainability.

5. Conclusions

Innovative has become imperative in the poultry industry for increasing productivity, monitoring environmental conditions, and automating poultry tasks for bird welfare. This paper presents a novel smart poultry management systems that uses hybrid smart sensors and Internet of Things technologies to effectively monitor temperature, relative humidity, ammonia gas level, lighting and other critical components of the poultry facility such as water consumption and PH level. The proposed system is connected with Bylnk software and features an interface that allows farmers to

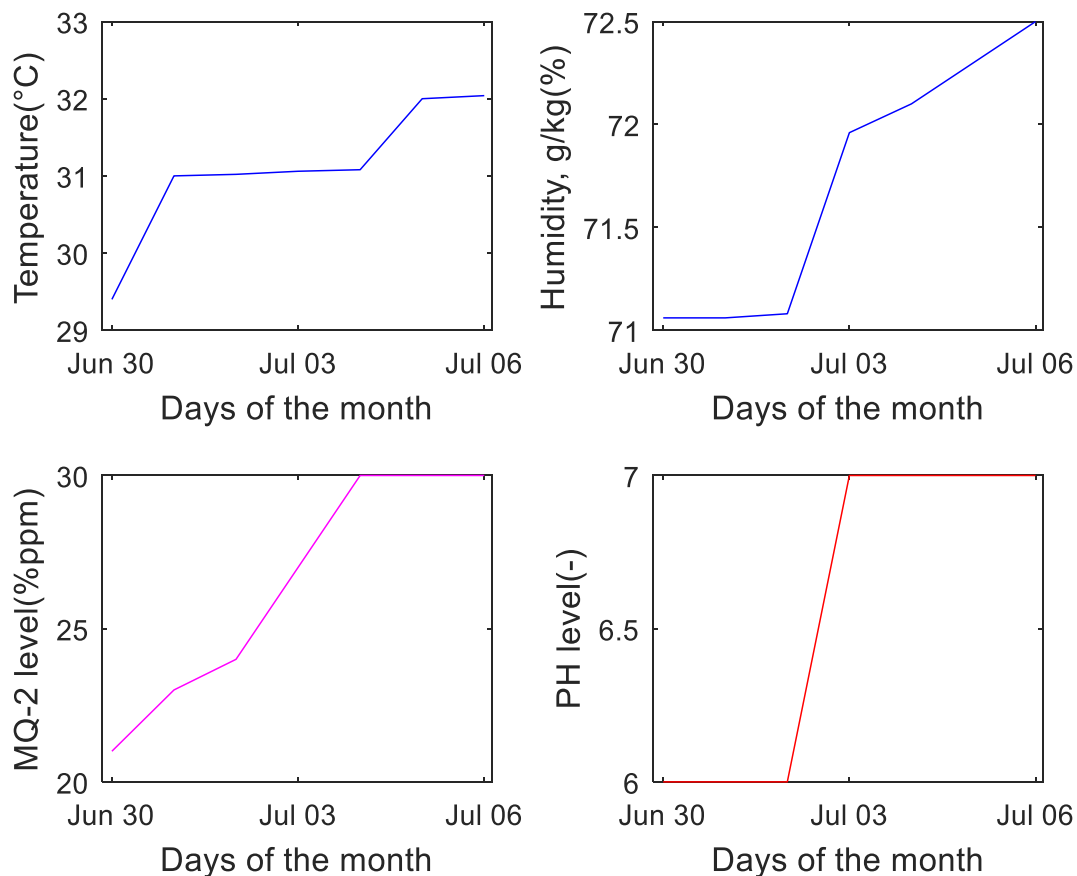


Fig. 11. Graphical display of (a) Temperature sensor (b) Relative humidity sensor (c) Ammonia gas sensor concentration (d) PH level.

Table 11. Performance comparison of the proposed system with existing studies.

No.	Paper	Network	Actuator	Technology	Physiological parameters	Addition features
1	Lufyagilia <i>et al</i> [114]	Wi-Fi	28.5	IoT	Temperature, Humidity and Ammonia gas	Web App
2	Orakwue <i>et al</i> [115]	Wi-Fi	27.8	IoT	Temperature, Humidity and Air quality	Motion detection
3	Batuto <i>et al</i> [116]	Wi-Fi	Servo motor, relay	IoT	Temperature, Humidity, Air quality	Google assistant
4	Mishra <i>et al</i> [117]	Wi-Fi	L293D, relay	IoT	Temperature and Humidity	GSM module
5	Syafar <i>et al</i> [118]	Wi-Fi	None	IoT	Temperature, Humidity and Air quality	Motion sensor
6	Fernando <i>et al</i> [119]	Wi-Fi	None	IoT, Web App	Temperature, Humidity, Ammonia gas and Luminosity	Web App
7	Proposed Scheme	Wi-Fi/Cellular (3G/4G/5G)	Electric fan, Electric heater, Humidifiers, Water pump, Extractor fan and Relay	IoT, Blynk cloud, GSM, Blynk mobile App	Temperature, Humidity, Lighting, Water level, PH level and Ammonia gas concentration	SMS, Alert notification, real-time data interface, lighting scheduling strategy, Water consumption control and

communicate with it. This module allows farmers to receive notifications, warming alerts, visualize real-time data, perform predictive analyses, and make informed chicken health decisions. Furthermore, the unique feature of the lighting management approach aids in managing the lighting conditions of the birds based on their age, resulting in increased egg production and optimal health. The recommended technique is critical for small-scale farmers in low-income countries like Nigeria, who have limited resources. Furthermore, the system has the ability to boost production while monitoring environmental conditions. The study aims to make a substantial contribution to some of the SDGs, such as improving nutrition and prioritizing sustainable agriculture in underdeveloped nations. As a result, transitioning to a smart poultry management system would not only improve poultry processes, but it would also significantly benefit small-scale farmers in Sub-Saharan Africa, particularly Nigeria, in terms of achieving food security for all by 2030 and encouraging precision agriculture. Although the study provided an innovative approach for poultry farming in Nigeria, future research into the integration of nitrogen sensors to reduce nitrogen content in poultry feed can improve the system. Additionally, feed sensors can be added into the system to improve poultry performance, as feed quality has a direct impact on bird performance [52]. Smart biosensors make it easier to detect pollutants and pathogens in

poultry products [120]. Disease detection sensors are another promising approach to preventing disease outbreaks in poultry systems.

Ethics information

This research complies with the ethical information for conducting scientific studies.

Source of Funding

This research did not receive any funding.

Conflict of Interest

The authors affirm that there is no conflict of interest regarding this work.

Data Availability

Data is available on request.

Author Contributions

A. A. Okubanjo: Conceptualization, Methodology, Experimentation, Writing- Original draft preparation, artworks design. I.K. Okakwu and N.S. Lawal reviewing the manuscript critically for important context A.A. Babalola. O.P. Alao and A. Olayiwola editing, and draft preparation.

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References

- [1] Spanaki K, Sivarajah U, Fakhimi M, Despoudi S, Irani Z. Disruptive technologies in agricultural operations: a systematic review of AI-driven AgriTech research. *Ann Oper Res* Jan. 2022;308(1–2):491–524.
- [2] Nnoli KP, Benyeogor MS, Olakanmi OO, Umanah DA. The computer farmer concept: human-cyberphysical systems for monitoring and improving agricultural productivity in Nigeria. In: *Proc. 2022 IEEE Niger. 4th Int. Conf. Disruptive Technol. Sustain. Dev. NIGERCON 2022*. 2; 2022.
- [3] Pawlak K, Kolodziejczak M. The role of agriculture in ensuring food security in developing countries: considerations in the context of the problem of sustainable food production. *Sustainability* Jul. 2020;12(13):5488.
- [4] Adeyanju Dolapo, Mburu John, Gituro Wainaina, Chumo Chepchumba, Mignouna Djana, Ogunniyi Adebayo, et al. Assessing food security among young farmers in Africa: evidence from Kenya, Nigeria, and Uganda. *Agric Food Econ* Feb. 2023;11(1):4.
- [5] Yimam HM, Cochrane L, Lemma MD. Not all crops are equal: the impacts of agricultural investment on job creation by crop type and investor type. *Heliyon* Jul. 2022;8(7):e09851.
- [6] Rafael BM. The importance of agricultural development projects: a focus on sustenance and employment creation in Kenya, Malawi, Namibia, Rwanda, and Uganda. *J Agric Chem Environ* 2023;12(2):152–70.
- [7] Adeyanju D, Mburu J, Gituro W, Chumo C, Mignouna D, Mulinganya N. Harnessing the job creation capacity of young rural agripreneurs: a quasi-experimental study of the ENABLE program in Africa. *Soc Sci Humanit Open* 2024; 9(June 2023):100791.
- [8] Sertoglu K, Ugural S, Bekun FV. International journal of economics and financial issues the contribution of agricultural sector on economic growth of Nigeria. *Int J Econ Financ Issues* 2017;7(1):547–52.
- [9] Serna-Loaiza S, Carmona-Garcia E, Cardona CA. Potential raw materials for biorefineries to ensure food security: the Cocoyam case. *Ind Crops Prod* Dec. 2018;126(September): 92–102.
- [10] Voronin BA, Chupina IP, Voronina YV, Kukhar VS, Simachkova NN. About agricultural products, raw materials and food with improved characteristics (scientific commentary on the Federal Law). *IOP Conf Ser Earth Environ Sci* Jan. 2022;949(1):012025.
- [11] Christiaensen L, Martin W. Agriculture, structural transformation and poverty reduction: eight new insights. *World Dev* Sep. 2018;109:413–6.
- [12] Gassner A, Harris D, Maus K, Terheggen A, Lopes C, Finlayson RF, et al. Poverty eradication and food security through agriculture in Africa: rethinking objectives and entry points. *Outlook Agric* Dec. 2019;48(4):309–15.
- [13] Otekunrin OA, Momoh S, Ayinde IA, Otekunrin OA. How far has Africa gone in achieving sustainable development goals? Exploring African dataset. *Data Brief* 2019;27: 104647.
- [14] Mottet A, Tempio G. Global poultry production: current state and future outlook and challenges. *Worlds Poult Sci J* 2017;73(2):245–56.
- [15] Morchid A, Jebabra R, Khalid HM, El Alami R, Qjidaa H, Ouazzani Jamil M. IoT-based smart irrigation management system to enhance agricultural water security using embedded systems, telemetry data, and cloud computing. *Results Eng* Sep. 2024;23:102829.
- [16] Ward AJ, Lewis DM, Green FB. Anaerobic digestion of algae biomass: a review. *Algal Res* 2014;5(1):204–14. Elsevier.
- [17] Morchid A, Muhammad Alblushi IG, Khalid HM, El Alami R, Sitaramanan SR, Muyeen SM. High-technology agriculture system to enhance food security: a concept of smart irrigation system using Internet of Things and cloud computing. *J Saudi Soc Agric Sci* November 2023, Feb. 2024; 23(8).
- [18] Rehman AU, Alamoudi Y, Khalid HM, Morchid A, Muyeen SM, Abdelaziz AY. Smart agriculture technology: an integrated framework of renewable energy resources, IoT-based energy management, and precision robotics. *Clean Energy Syst* Dec. 2024;9(July):100132.
- [19] Said Z, Sharma Prabhakar, Thi Bich Nhuong Quach, Bora Bhaskor J, Lichtfouse Eric, Khalid Haris M, et al. Intelligent approaches for sustainable management and valorisation of food waste. *Bioresour Technol* Jun. 2023;377: 128952.
- [20] Morchid A, El Alami R, Raedah AA, Sabbar Y. Applications of internet of things (IoT) and sensors technology to increase food security and agricultural Sustainability: benefits and challenges. *Ain Shams Eng J* 2024;15(3):102509.
- [21] Zhivkov T, Sklar EL, Botting D, Pearson S. 5G on the farm: evaluating wireless network capabilities and needs for agricultural robotics. *Machines* Nov. 2023;11(12):1064.
- [22] Li X, Zhang J, Jin W, Liu W. Key technology implementation of poultry breeding system for 5G intelligent IOT. In: *2020 IEEE conference on telecommunications, optics and computer science (TOCS)*; 2020. p. 372–6.
- [23] Alindekon S, Rodenburg TB, Langbein J, Puppe B, Wilmsmeier O, Louton H. Setting the stage to tag 'n' track: a guideline for implementing, validating and reporting a radio frequency identification system for monitoring resource visit behavior in poultry. *Poult Sci* Aug. 2023; 102(8):102799.
- [24] Tang R, Aridas NK, Abu Talip MS. Design of wireless sensor network for agricultural greenhouse based on improved zigbee protocol. *Agriculture* Jul. 2023;13(8):1518.
- [25] Okinda Cedric, Nyalala Innocent, Korohou Tchalla, Okinda Celestine, Wang Jintao, Achieng Tracy, et al. A review on computer vision systems in monitoring of poultry: a welfare perspective. *Artif Intell Agric* 2020;4:184–208.
- [26] Campbell Mairead, Miller Paul, Diaz-Chito Katherine, Hong Xin, McLaughlin Niall, Parvinzamin Farzad, et al. A computer vision approach to monitor activity in commercial broiler chickens using trajectory-based clustering analysis. *Comput Electron Agric* Feb. 2024;217(February 2023): 108591.
- [27] Sidarto LP, Hamka A. Improving halal traceability process in the poultry industry utilizing blockchain technology: use case in Indonesia. *Front Blockchain* December 2021;4:1–8.
- [28] Ren G, Lin T, Ying Y, Chowdhary G, Ting KC. Agricultural robotics research applicable to poultry production: a review. *Comput Electron Agric* Feb. 2020;169(June 2019):105216.
- [29] van der Merwe D, Burchfield DR, Witt TD, Price KP, Sharda A. Drones in agriculture. In: *Advances in agronomy*. 1st ed.vol. 162. Elsevier Inc.; 2020. p. 1–30.
- [30] Pereira WF, Fonseca L da S, Putti FF, Góes BC, Naves L de P. Environmental monitoring in a poultry farm using an instrument developed with the internet of things concept. *Comput Electron Agric* Mar. 2020;170(November 2019):105257.
- [31] Sakamoto KS, Benincasa NC, Silva IJO, Lobos CMV. The challenges of animal welfare in modern Brazilian poultry farming. *J Anim Behav Biometeorol* 2020;8(2):131–5.
- [32] Göransson L, Lundmark Hedman F. The perks of being an organic chicken – animal welfare science on the key features of organic poultry production. *Front Anim Sci* May 2024;5(May):1–15.
- [33] An Y, Xing H, Zhang Y, Jia P, Gu X, Teng X. The evaluation of potential immunotoxicity induced by environmental pollutant ammonia in broilers. *Poult Sci* Aug. 2019;98(8): 3165–75.

- [34] Morchid A, Muhammad Alblushi IG, Khalid HM, El Alami R, Sitaramanan SR, Muyeen SM. High-technology agriculture system to enhance food security: a concept of smart irrigation system using Internet of Things and cloud computing. *J Saudi Soc Agric Sci* November 2023, 2024;23(8).
- [35] Bumanis N, Kviesis A, Tjukova A, Arhipova I, Paura L, Vitols G. Smart poultry management platform with egg production forecast capabilities. *Procedia Comput Sci* 2022; 217(2022):339–47.
- [36] Sanjaya DD, Fadlil A. Monitoring temperature and humidity of boiler chicken cages based on internet of things (IoT). *Bul Ilm Sarj Tek Elektro May* 2023;5(2):180–9.
- [37] Olanubi OO, Akano TT, Asaolu OS. Design and development of an IoT - based intelligent water quality management system for aquaculture. *J Electr Syst Inf Technol* 2024; 11.
- [38] Cassuce DC, Zolnier S, Cecon PR, Vieira MDEFA. Atualização da temperaturas de conforto térmico para frangos de corte de até 21 dias de idade. *Eng Agríc* 2013; 33(1):28–36.
- [39] Kim D-H, Lee Y-K, Kim S-H, Lee K-W. The impact of temperature and humidity on the performance and physiology of laying hens. *Animals* Dec. 2020;11(1):56.
- [40] Liu ZL, Chen Y, Xue JJ, Huang XF, Chen ZP, Wang QG, et al. Effects of ambient temperature on the growth performance, fat deposition, and intestinal morphology of geese from 28 to 49 days of age. *Poult Sci* May 2022;101(5): 101814.
- [41] Sitka A, Szulc P, Smykowski D, Jodkowski W. Application of poultry manure as an energy resource by its gasification in a prototype rotary counterflow gasifier. *Renew Energy* Sep. 2021;175:422–9.
- [42] Price KR, Guerin MT, Barta JR. Success and failure: the role of relative humidity levels and environmental management in liveEimeria vaccination of cage-reared replacement layer pullets. *J Appl Poultry Res* Sep. 2014;23(3):523–35.
- [43] Oluwagbenga EM, Fraley GS. Heat stress and poultry production: a comprehensive review. *Poult Sci* 2023;102(12): 103141.
- [44] Soliman AS, Khafaga MA, Soliman FN, El-Sabrout KM. Effect of different lighting sources on the performance of broiler breeder hens. *J Anim Behav Biometeorol* Jul. 2023; 11(3):e2023026.
- [45] El-Sabrout K, El-Deek A, Ahmad S, Usman M, Dantas MRT, Souza-Junior JBF. Lighting, density, and dietary strategies to improve poultry behavior, health, and production. *J Anim Behav Biometeorol* 2022;10(1):1–17.
- [46] Purswell JL, Olanrewaju HA, Linhoss JE. Effect of light intensity adjusted for species-specific spectral sensitivity on live performance and processing yield of male broiler chickens. *J Appl Poultry Res* Dec. 2018;27(4):570–6.
- [47] Raccoursier M, Thaxton YV, Christensen K, Aldridge DJ, Scanes CG. Light intensity preferences of broiler chickens: implications for welfare. *Animal* 2019;13(12):2857–63.
- [48] Bahuti M, Yanagi Junior T, Fassani EJ, Ribeiro BPVB, de Lima RR, Campos AT. Evaluation of different light intensities on the well-being, productivity, and eggs quality of laying hens. *Comput Electron Agric* Dec. 2023; 215(Decemer):108423.
- [49] Takeshima K, Hanlon C, Sparling B, Korver DR, Bédécarrats GY. Spectrum lighting during pullet rearing and its impact on subsequent production performance in layers. *J Appl Poultry Res* Dec. 2019;28(4):1262–78.
- [50] Leeson S, Caston LJ, Summers JD. Performance of layers given two-hour midnight lighting as growing pullets. *J Appl Poultry Res* Oct. 2003;12(3):313–20.
- [51] Nissa Shaista S, Sheikh Islam U, Altaie Hayman AA, Adil Sheikh, Banday Mohammad T, Kamal Mahmoud, et al. Impacts of various lighting programs on chicken production and behavior – a review. *Ann Anim Sci* Oct. 2024;24(4): 1065–79.
- [52] Swelum Ayman A, El-Saadony Mohamed T, Abd El-Hack Mohamed E, Abo Ghanima Mahmoud M, Shukry Mustafa, Alhotan Rashed A, et al. Ammonia emissions in poultry houses and microbial nitrification as a promising reduction strategy. *Sci Total Environ* 2021;781(2): 146978. Elsevier B.V.
- [53] Sheikh IU. Ammonia production in the poultry houses and its harmful effects. *Worlds Poult Sci J* 2018;40(2): 99–113.
- [54] Mohammad Al-Kerwi MS, Mardenli O, Mahdi Jasim MR, Al-Majeed MA. Effects of harmful gases emitted from poultry houses on productive and health performance. *IOP Conf Ser Earth Environ Sci* Jul. 2022;1060(1):012082.
- [55] Bist RB, Subedi S, Chai L, Yang X. Ammonia emissions, impacts, and mitigation strategies for poultry production: a critical review. *J Environ Manag* Feb. 2023;328:116919.
- [56] El Sabry MI, Romeih ZU, Stino FKR, Khosht AR, Aggrey SE. Water scarcity can be a critical limitation for the poultry industry. *Trop Anim Health Prod* Jun. 2023;55(3):215.
- [57] Di Martino G, Piccirillo A, Giacomelli M, Comin D, Gallina A, Capello K, et al. Microbiological, chemical and physical quality of drinking water for commercial turkeys: a cross-sectional study. *Poult Sci* 2018;97(8):2880–6.
- [58] Dirk Van Der Klis J, De Lange L. Water intake of poultry. *Proc 19th Eur Symp Poult Nutr* 2013;(July):1–10.
- [59] Ojo RO, Ajayi AO, Owolabi HA, Oyedele LO, Akanbi LA. Internet of Things and Machine Learning techniques in poultry health and welfare management: a systematic literature review. *Comput Electron Agric* Sep. 2022;200(1): 107266.
- [60] Oluwagbenga EM, Fraley GS. Heat stress and poultry production: a comprehensive review. *Poult Sci* Dec. 2023; 102(12):103141.
- [61] Ahmed G, Malick RAS, Akhunzada A, Zahid S, Sagri MR, Gani A. An approach towards IoT-based predictive service for early detection of diseases in poultry chickens. *Sustainability* Dec. 2021;13(23):13396.
- [62] Ojo RO, Ajayi AO, Owolabi HA, Oyedele LO, Akanbi LA. Internet of Things and Machine Learning techniques in poultry health and welfare management: a systematic literature review. *Comput Electron Agric* Sep. 2022;200(July 2022):107266.
- [63] Ojo RO, Ajayi AO, Owolabi HA, Oyedele LO, Akanbi LA. Internet of Things and Machine Learning techniques in poultry health and welfare management: a systematic literature review. *Comput Electron Agric* Sep. 2022;200(75): 107266.
- [64] Xu J, Gu B, Tian G. Review of agricultural IoT technology. *Artif Intell Agric* 2022;6:10–22.
- [65] Ojo RO, Ajayi AO, Owolabi HA, Oyedele LO, Akanbi LA. Internet of Things and Machine Learning techniques in poultry health and welfare management: a systematic literature review. *Comput Electron Agric* 2022;200(July): 107266.
- [66] Onibonije MO. IoT-based synergistic approach for poultry management system. In: 2021 IEEE international IOT, electronics and mechatronics conference (IEMTRONICS); 2021. p. 1–5.
- [67] Ahmed MM, Hassanien EE, Hassanien AE. A smart IoT-based monitoring system in poultry farms using chicken behavioural analysis. *Internet of Things (Netherlands)* 2024; 25(November 2023):101010.
- [68] Subeesh A, Mehta CR. Automation and digitization of agriculture using artificial intelligence and internet of things. *Artif Intell Agric* 2021;5:278–91.
- [69] Prabowo MCA, Sayekti I, Astuti S, Nursaputro ST, Supriyati S. Development of an IoT-based egg incubator with PID control system and web application. *JOIV Int J Informatics Vis Mar.* 2024;8(1):465.
- [70] Nicolas RDM, Zhou WS, Kitamura SC, Samonte MJC. An IoT monitoring assistant for chicken layer farms. In: 2019 international conference on information and communication technology convergence (ICTC); 2019. p. 71–5.
- [71] Nalendra AK, Priyawaspada H, Nur Fuad M, Mujiono M, Wahyudi D. Monitoring system IoT-broiler chicken cage

- effectiveness of seeing reactions from chickens. *J Phys Conf Ser Jun.* 2021;1933(1):012097.
- [72] Bose R, Roy S, Mondal H. A novel algorithmic electric power saver strategies for real-time smart poultry farming. *e-Prime - Adv Electr Eng Electron Energy* 2022;2(July): 100053.
- [73] Toppel K, Kaufmann F, Schön H, Gauly M, Andersson R. Effect of pH-lowering litter amendment on animal-based welfare indicators and litter quality in a European commercial broiler husbandry. *Poult Sci Mar.* 2019;98(3):1181–9.
- [74] Chigwada J, Mazunga F, Nyamhere C, Mazheke V, Taruvinga N. Remote poultry management system for small to medium scale producers using IoT. *Sci African Nov.* 2022; 18:e01398.
- [75] Astill J, Dara RA, Fraser EDG, Roberts B, Sharif S. Smart poultry management: smart sensors, big data, and the internet of things. *Comput Electron Agric* December 2019; 170:105291. Mar. 2020.
- [76] Mandal S, Yadav A, Panme FA, Devi KM, Kumar S.M. S. Adaption of smart applications in agriculture to enhance production. *Smart agricultural technology*, vol. 7. February. Elsevier B.V.; 2024. p. 100431.
- [77] Rajak P, Ganguly A, Adhikary S, Bhattacharya S. Internet of Things and smart sensors in agriculture: scopes and challenges. *J Agric Food Res Dec.* 2023;14(September):100776.
- [78] Quasim MT, Nisa Khair ul, Khan Mohammad Zunnun, Husain Mohammad Shahid, Alam Shadab, Shuaib Mohammed, et al. An internet of things enabled machine learning model for Energy Theft Prevention System (ETPS) in Smart Cities. *J Cloud Comput Nov.* 2023; 12(1):158.
- [79] Milosevic B, Ciric S, Lalic N, Milanovic V, Savic Z, Omerovic I, et al. Machine learning application in growth and health prediction of broiler chickens. *Worlds Poult Sci J Sep.* 2019;75(3):401–10.
- [80] Ali Muhammad, Imran Muhammad, Baig Muhammad Shamim, Shah Adil, Ullah Syed Sajid, Alroobaea Roobaea, et al. Intelligent control shed poultry farm system incorporating with machine learning. *IEEE Access* 2024;12(April): 58168–80.
- [81] Gandhi K I. AIoT-driven edge computing for rural small-scale poultry farming: smart environmental monitoring and anomaly detection for enhanced productivity. *Int J Recent Innov Trends Comput Commun Sep.* 2023;11(8):44–52.
- [82] Debauche O, Mahmoudi S, Mahmoudi SA, Manneback P, Bindelle J, Lebeau F. Edge computing and artificial intelligence for real-time poultry monitoring. *Procedia Comput Sci* 2020;175(August):534–41.
- [83] Reham A Hosny MFA, Alatfeehy Nayera M, Resistance CA, Hosny RA, Alatfeehy NM, Abdelaty MF. Application of artificial intelligence in the management of poultry farms and combating antimicrobial resistance. *Egypt J Anim Heal Jul.* 2023;3(3):91–102.
- [84] Neethirajan S. Artificial intelligence and sensor innovations: enhancing livestock welfare with a human-centric approach. *Human-Centric Intell Syst Nov.* 2023;4(1):77–92.
- [85] Malini T, Aswath DL, Abhishek R, Kirubhakaran R, Anandhamurugan S. IoT based smart poultry farm monitoring. In: 2023 9th International Conference on Advanced Computing and Communication Systems (ICACCS). 4; 2023. p. 13–8. 2.
- [86] Odilov BA, Madraimov, Askariy, Yusupov, Otabel Y, Karimov, Nodir R, Alimova, Rakhima, Yakhshieva, Zukhra Z, et al. Utilizing deep learning and the internet of things to monitor the health of aquatic ecosystems to conserve biodiversity. *Nat Eng Sci May* 2024;9(1):72–83.
- [87] Singh M, Kumar R, Tandon D, Sood P, Sharma M. Artificial intelligence and IoT based monitoring of poultry health: a review. In: 2020 IEEE international conference on communication, networks and satellite (Comnetsat); 2020. p. 50–4.
- [88] Manshor N, Abdul Rahiman AR, Yazed MK. IoT based poultry house monitoring. In: 2019 2nd international conference on communication engineering and technology (ICCET); 2019. p. 72–5. February.
- [89] Chang Y-S, Tu L-Y. Use of IoT sensors to build an intelligent monitoring and control system for poultry house environment. *Sensor Mater Dec.* 2023;35(12):3997.
- [90] Elmoulart M, Debauche O, Mahmoudi S, Mahmoudi SA, Manneback P, Lebeau F. Edge computing and artificial intelligence for landslides monitoring. *Procedia Comput Sci* 2020;177(2019):480–7.
- [91] Jebari H, Hayani Mechkouri M, Rekiek S, Reklouai K. Poultry-edge-AI-IoT system for real-time monitoring and predicting by using artificial intelligence. *Int J Interact Mob Technol Jun.* 2023;17(12):149–70.
- [92] Choosumrong S, Raghavan V, Pothong T. Smart poultry farm based on the real-time environment monitoring system using internet of things. *Naresuan Agric J* 2019;16(2): 18–26.
- [93] Phiri H, Kunda D, Phiri J. An IoT smart broiler farming model for low income farmers. *Int J Recent Contrib from Eng Sci IT Nov.* 2018;6(3):95.
- [94] So-In C, Poolsanguan S, Rujirakul K. A hybrid mobile environmental and population density management system for smart poultry farms. *Comput Electron Agric* 2014;109: 287–301.
- [95] Zheng H, Zhang T, Fang C, Zeng J. Design and implementation of poultry farming information management system based on cloud database. 2021.
- [96] Sasirekha R, Kaviya R, Saranya G, Mohamed A, Iroda U. Smart poultry house monitoring system using IoT. *E3S Web Conf* 2023;399.
- [97] Corkery G, Ward S, Kenny C, Hemmingway P. Incorporating smart sensing technologies into the poultry industry. *J World's Poult Res (JWPR)* 2013;3(4):106–28.
- [98] Cruz E, Hidalgo-Rodriguez M, Acosta-Reyes AM, Rangel JC, Boniche K. AI-based monitoring for enhanced poultry flock management. *Agric* 2024;14(12):1–26.
- [99] K C K, Subedi K, Sharma S, Paneru P. IoT based smart poultry management system. *J ISMAC Mar.* 2024;6(1): 39–53.
- [100] Vinueza-Naranjo PG, Nascimento-Silva HA, Rumipamba-Zambrano R, Ruiz-Gomes I, Rivas-Lalaleo D, Patil NJ. IoT-based smart agriculture and poultry farms for environmental sustainability and development. May. 2022.
- [101] Rajakumar G, Jenifer GG, Kumar TA, Samuel TSA. Design of an energy efficient IoT system for poultry farm management. Singapore: Springer Nature; 2022.
- [102] Lashari MH, Karim S, Alhussein M, Hoshu AA, Aurangzeb K, Anwar MS. Internet of Things-based sustainable environment management for large indoor facilities. *PeerJ Comput Sci* 2023;9.
- [103] Andre Rafaela S, Mercante Luiza A, Facure Murilo HM, Sanfelice Rafaela C, Fugikawa-Santos Lucas, Swager Timothy M, et al. Recent progress in amine gas sensors for food quality monitoring: novel architectures for sensing materials and systems. *ACS Sens* 2022;7(8): 2104–31.
- [104] Kumar Mohanty D, Kondala Rao T, S HK, Agme R, Gogoi C, Velu CM. IoT applications for livestock management and health monitoring in modern farming. *Theory Into Pract* 2024;2024(4):2141–53.
- [105] Suberu MY, Bashir N, Mustafa MW. Biogenic waste methane emissions and methane optimization for bio-electricity in Nigeria. *Renew Sustain Energy Rev* 2013;25: 643–54. Elsevier Ltd.
- [106] Issa AA, Majed S, Ameer A, Al-Jawahry HM. IoT and AI in livestock management: a game changer for farmers. *E3S Web Conf* 2024;491.
- [107] Isaac JO. Iot - livestock monitoring and management system. *Int J Eng Appl Sci Technol* 2021;5(9):254–7.
- [108] Nandini B, Mawale KS, Giridhar P. Nanomaterials in agriculture for plant health and food safety: a comprehensive review on the current state of agro-nanoscience. *3 Biotech* 2023;13(3):1–21.

- [109] Chhipa H. Applications of nanotechnology in agriculture. 1st ed.vol. 46. Elsevier Ltd.; 2019.
- [110] Taştan M. IoT based wearable smart health monitoring system. Celal Bayar Univ J Sci Sep. 2018;14(3):343–50.
- [111] Nagar A, Pradeep T. Clean water through nanotechnology: needs, gaps, and fulfillment. ACS Nano 2020;14(6):6420–35.
- [112] Otles S, Sahyar BY. Nanotechnology application and emergence in agriculture. Emerg Trends Agri-nanotech Fundam Appl Asp 2018;2(6):204–13.
- [113] Feddes JJR, Emmanuel EJ, Zuidhof MJ. Broiler performance, body weight variance, feed and water intake, and carcass quality at different stocking densities. Poult Sci 2002;81(6):774–9.
- [114] Lufyagila B, Machuve D, Clemen T. IoT-powered system for environmental conditions monitoring in poultry house: a case of Tanzania. African J Sci Technol Innov Dev Jun. 2022; 14(4):1020–31.
- [115] Orakwue SI, Al-Khafaji HMR, Chabuk MZ. IoT based smart monitoring system for efficient poultry farming. Webology Jan. 2022;19(1):4105–12.
- [116] Batuto A, Dejeron TB, Dela Cruz P, Samonte MJC. E-poultry: an IoT poultry management system for small farms. In: 2020 IEEE 7th international conference on industrial engineering and applications (ICIEA); 2020. p. 738–42.
- [117] Mishra S, Sheikh A, Chore S, Kshirsagar S. IoT based automatic poultry feeding and smart poultry farm system. Int Conf Innov Eng Technol Sci Manag 2019;9(5): 33–6.
- [118] Ali Muhammad, Imran Muhammad, Baig Muhammad Shamim, Shah Adil, Ullah Syed Sajid, Alroobaea Roobaea, et al. Intelligent control shed poultry farm system incorporating with machine learning. IEEE Access 2024;12(10): 58168–80.
- [119] Tartan EO, Ciflikli C. An android application for geo-location based health monitoring, consultancy and alarm system. Proc - Int Comput Softw Appl Conf 2018;2:341–4.
- [120] Renuga Devi N, Suganya T, Vignesh S, Rathish RJ, Nguyen TA, Rajendran S. Animal health monitoring using nanosensor networks. Nanosensors for Smart Agriculture 2022;2022:573–608. Elsevier.